

Isaiah Flager-Hearan

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EDUCATION

Southern Illinois University Edwardsville
Bachelor of Science, Computer Science

Dec 2024
Edwardsville, Illinois

WORK EXPERIENCE

National Great Rivers Research and Education Center

Jun 2024 - Aug 2024

Software Engineer Intern

Alton, Illinois

- Collaborated with a cross-functional team of four developers to build a responsive web application using Next.js and Tailwind CSS that visualizes real-time water-monitoring data from Illinois, Missouri, and Wisconsin, allowing stakeholders to access up-to-date environmental metrics
- Designed and implemented an optimized PostgreSQL database architecture for real-time water quality visualization and long-term scientific data storage, improving query performance by 20%
- Built a user-friendly data administration tool in Python, enabling non-technical scientists to manage database content independently without SQL knowledge
- Deployed full-stack application on DigitalOcean, integrating real-time data ingestion pipelines from 8 IoT monitoring stations across 3 states and maintaining 99.9% uptime for critical environmental reporting
- Developed interactive geospatial visualizations using React.js and Leaflet.js with dynamic map overlays for monitoring station data, enabling real-time environmental awareness across a multi-state region

SKILLS

- **Programming Languages:** Python, TypeScript, Java, Go
- **Backend:** FastAPI, Django, Spring Boot, Node.js, REST APIs, NumPy, Pandas, SQLAlchemy
- **Frontend:** React.js, Next.js, Tailwind CSS, HTML
- **Databases:** PostgreSQL, MongoDB, TimescaleDB, Redis
- **Cloud & DevOps:** AWS, GCP, Docker, DigitalOcean, Supabase, GitHub Actions, CI/CD, Nginx
- **AI/ML:** PyTorch, OpenAI API
- **Tools:** Git, Linux, Sentry, Agile/Scrum

PROJECTS

EdgarFeed | [GitHub](#)

Jan 2026 - Present

- Built a polyglot microservices platform that ingests live SEC Form 4 insider trading filings and surfaces a ranked feed of statistically notable executive transactions
- Engineered an AWS Lambda (Python) ingestion pipeline parsing government XML from EDGAR, with idempotent writes and EventBridge cron triggering daily at market close
- Implemented event study methodology in a FastAPI microservice computing Cumulative Abnormal Return via OLS regression, and trained an XGBoost anomaly scoring model achieving AUC > 0.75 on 2023–2024 holdout data
- Deployed on AWS with EC2, RDS, Lambda, and S3 + CloudFront; automated full ingestion pipeline from SEC filing to ranked feed

Polywatch | [polywatch.dev](#)

Dec 2025 - Present

- Built real-time data pipeline collecting trading data from 600+ Polymarket prediction markets, processing 750K+ time-series snapshots with automated 5-minute collection cycles
- Implemented statistical pattern detection engine using z-score analysis and sliding window algorithms to identify volume spikes and price momentum anomalies with confidence scoring
- Designed FastAPI backend with SQLAlchemy ORM, TimescaleDB hypertables for time-series optimized PostgreSQL storage, Alembic migrations, and signal deduplication logic for efficient historical querying
- Deployed on Google Cloud Platform using Docker containers on Cloud Run for the API, Cloud Run Jobs with Cloud Scheduler for data collection, Compute Engine for TimescaleDB, and Cloud Storage for static frontend hosting

Steam Price Tracker | [steampricetracker.com](#)

May 2025 - Dec 2025

- Built full-stack price tracking platform with 300+ registered users, enabling Steam gamers to monitor prices across thousands of games and receive automated email alerts on discounts via Mailjet
- Designed a recommendation service leveraging OpenAI APIs to generate personalized game suggestions based on user preferences, tracked titles, and genre metadata
- Architected secure authentication with hybrid JWT system (Bearer tokens + HttpOnly cookies) and integrated Stripe payment processing for donations with automatic receipt generation
- Deployed production backend on AWS EC2 with Nginx reverse proxy and Cloudflare DNS, with automated hourly price syncing via APScheduler background jobs